OAKLAND COMMUNITY COLLEGE®
ROBOTICS/AUTOMATED SYSTEMS TECHNOLOGY

ROB 2140 FINAL

NO FLASH DRIVES ARE ALLOWED
IF A FLASH DRIVE IS FOUND IN THE COMPUTER A ZERO GRADE WILL BE ISSUED.

TRANSFER ANY FILES BEFORE BEGINNING THE FINAL.
REMOVE THE FLASH DRIVE FROM THE COMPUTER AFTER TRANSFERRING.

SAVE BEFORE DOWNLOADING.

IF SYSTEM FAILS AND YOU DID NOT SAVE, YOU MAY LOSE YOUR WORK AND WILL HAVE TO REDO.

ALLOTTED TIME:
3 HOURS AND 45 MINUTES FROM THE COURSE SCHEDULED START TIME.

Files are in the folder, **STUDIO 5000**

REMEMBER TO PRINT TO OneNote

The following problems uses STUDIO 5000, STUDIO Emulate 5000, and FactoryTalk View ME

Final problem Dispense Tank must use the filename: DISPENSE (25 Points)

Final problem Torque Motors must use the filename: TORQUE_FINAL. (25 Points)

Files are in the folder, **SIEMENS**

REMEMBER TO PRINT TO OneNote

The following problem uses Siemens TIA

Final problem Press Jam must use the filename: JAM. (25 Points)

The following problem uses FactoryTalk View Studio ME

DO NOT PRINT, HMI WILL BE CHECKED ON THE COMPUTER

Final problem HMI uses the filename: TORQUE FINAL *session* (PLC: 25 Points)

IF YOU CANNOT FIND THE FILES LOCATION, THE FILE WAS NOT LOADED ON THE COMPUTER, OR THE FILE FAILS TO OPEN, NOTIFY THE INSTRUCTOR IMMEDIATELY. IF YOU FAIL TO NOTIFY THE INSTRUCTOR OF ANY TECHNICAL ISSUE, IT DOES NOT CONSTITUTE A REASON FOR NOT COMPLETING THE FINAL PROBLEM.

HMI SCREENS

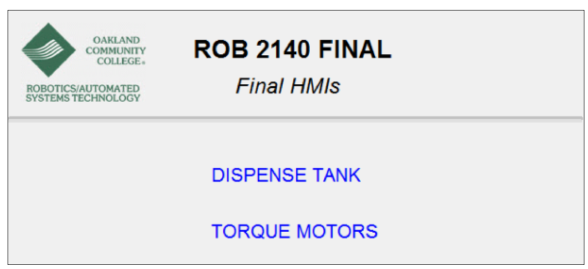
STUDIO 5000 Dispense Tank and Torque Motors

Open **FactoryTalk View ME Station**.

The HMI is a runtime file, see HMI Procedures, section Station Runtime Application to start the **FINAL_HMI** file.

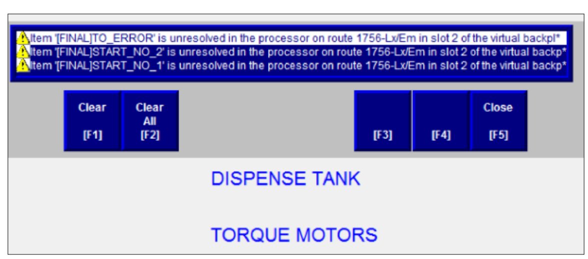
A Program Selection menu appears:

- **Do Not** open a application HMI screen until the STUDIO 5000 program has been downloaded and is in RUN mode.
- **Return to the Program Selection screen** before changing the STUDIO 5000 processor to PROGRAM mode or OFFLINE.



Each program HMI screen has a link back the Program Selection screen.

After downloading the STUDIO 5000 program and placing in **RUN** mode, an alarm screen will appear because the processor went offline to download. Press **F5.**



SIEMENS Press Jam

No HMI is used.

STUDIO 5000 APPLICATION - DISPENSE TANK

REMEMBER TO PRINT TO OneNote FINAL FOLDER
See desktop document <u>Printing Add-On to OneNote</u>

STUDIO 5000 PROGRAM NAME: DISPENSE

Create the logic to control a vat that will mixing of a solid and liquid.

Programming Suggestion:

Use the compare instructions with rungs that have internal to Binary Data Type outputs to control the "real world" outputs.

SEQUENCE OF OPERATIONS

1. The sensors for the Solid Load Cell and Fluid Flow Meter are networked, transmitting the Raw Bit values, ADD-ON instructions must be used to scale the values to the Engineering Units.

2. The operator will press and release a momentary START pushbutton.

3. An output for a SOLID MATERIAL DISPENSE GATE will be energized, allowing the material to go onto a WEIGHING PLATFORM. When 35 or more pounds have been dispensed, the SOLID MATERIAL DISPENSE GATE output will turn off.

4. The signal that 35 pounds is on the WEIGHTING PLATFORM will trigger three output actions that will simultaneously occur:

   - A DISCHARGE AUGER will turn on to move the solid material into the vat.
   - An output for the DISPENSE LIQUID VALVE will turn on.
   - The MIXER MOTOR will turn on.

5. When the weight senor at the weighing platform reads zero, the DISCHARGE AUGER will turn off.

6. When 70 gallons of liquid have been dispensed, the DISPENSE LIQUID VALVE will turn off.

   NOTE: The FLOW SENSOR CONTROLLER sets the meter to zero when the Fluid Valve is Off and the meter's reading is at the setpoint.

7. The MIXER MOTOR will run for an additional 10 second after the vat is filled with the specified amount of fluid.

8. When the MIXER MOTOR turns off, a DISCHARGE MIXTURE VALVE will turn on.

9. When the mixture has drained out of the vat, the LOW LEVEL FLOAT SWITCH will signal the DISCHARGE MIXTURE VALVE to turn off.

I/O CONFIGURATION:

Slot 2 – STUDIO Emulate 5000 Controller
Slot 3 - 1756-IB16
Slot 4 - 1756-OB16E
Slot 5 - 1756-ENET/B, Ethernet/IP to sensors

INPUTS:

Address	Controller Tag	Description
Local:3:I.Data.0	START_PB	Start Momentary Normally Open Pushbutton
Local:3:I.Data.1	LOW_LEVEL_SW	Low Level Float Switch, Normally Closed

OUTPUTS:

Address	Controller Tag	Description
Local:4:O.Data.0	DISPENSE_GATE	Solid Material Dispense Gate
Local:4:O.Data.1	DISCHARGE_AUGUR	Solid Material Discharge Augur
Local:4:O.Data.2	FLUID_VALVE	Dispense Liquid Valve
Local:4:O.Data.3	MIXER_MOTOR	Mixer Motor
Local:4:O.Data.4	DISCHARGE_VALVE	Discharge Mixture Valve

ANALOG INPUTS:

Address	Controller Tag	Description
SENSOR:I.Data[0]	LOAD_CELL	Analog Bit Value from Weight Sensor
SENSOR:I.Data[1]	FLOW_METER	Analog Bit Value from Flow Sensor

INTERNALS:

Controller Tag	Data Type	Description
SOLID (NOTE)	ANALOG_IN	Add On Instruction for Solid
SOLID_WEIGHT	DINT	Scaled in EU Solid Value
FLUID (NOTE)	ANALOG_IN	Add On Instruction for Fluid
FLUID_VOL	DINT	Scaled in EU Fluid Value

NOTE: Add-On instruction must be created first before entering these tags.

TIMERS:

(1) The tags for the status bits DN, TT, and EN are created automatically when timer tag is created.

Param/Local Tag	Data Type	Description
MIX_DWELL	Timer	Mixer Motor Timer

INTERNALS:

Param/Local Tag	Data Type	Description
SOLID_FILLED	BOOL	At Weighing Platform
SOLID_EMPTY	BOOL	At Weighing Platform
FLUID_FILLED	BOOL	Dispensed from Liquid Valve
LOAD_CELL_MAX	DINT	Maximum Solid Setting
LOAD_CELL_MIN	DINT	Minimum Solid Setting
FLOWMETER_MAX	DINT	Maximum Fluid Setting
FLOWMETER_MIN	DINT	Minimum Fluid Setting
RAW_MAX	DINT	Maximum Bit Value of Sensors
RAW_MIN	DINT	Minimum Bit Value of Sensors

SEE NEXT PAGE FOR ADD ON INSTRUCTION

ADD-ON INSTRUCTION

INSTRUCTION NAME: ANALOG_IN

PROGRAMMING LOGIC: Either FUNCTION BLOCK DIAGRAM or STRUCTURED TEXT.

If using Structured Text, see the document Studio 5000 Analog Scale*, page 3, for an example of creating the Tag for the* ***SCL*** *instruction.*

PARAMETERS:

Name	Usage	Data Type	Required and Visible	Description
Raw_Max	Input	REAL	✓	Maximum Bit Value
Raw_Min	Input	REAL	✓	Minimum Bit Value
EU_Max	Input	REAL	✓	Engineering Unit Maximum
EU_Min	Input	REAL	✓	Engineering Unit Minimum
Sensor	Input	REAL	✓	Analog Channel, Source
EU_Value	Output	REAL	✓	Scaled Engineering Unit Value

READ BEFORE RUNNING PROGRAM AND HMI

When the START pushbutton is pressed the DISPENSE_GATE value turns ON.

The SOLID_WEIGHT value will begin to increase.

When the SOLID_WEIGHT value is at 35:
DISPENSE_GATE value turns OFF.
DISCHARGE_AUGUR value turns ON,
FLUID_VALVE value turns ON, and
MIXER_MOTOR value turns ON.

The SOLID_WEIGHT value will begin to decrease.

FLUID_VOL value will begin to increase.

At a 0 value for SOLID_WEIGHT, the DISCHARGE_AUGUR turns OFF.

At a value of 70 for the FLUID_VOL, the FLUID_VALVE turns OFF.

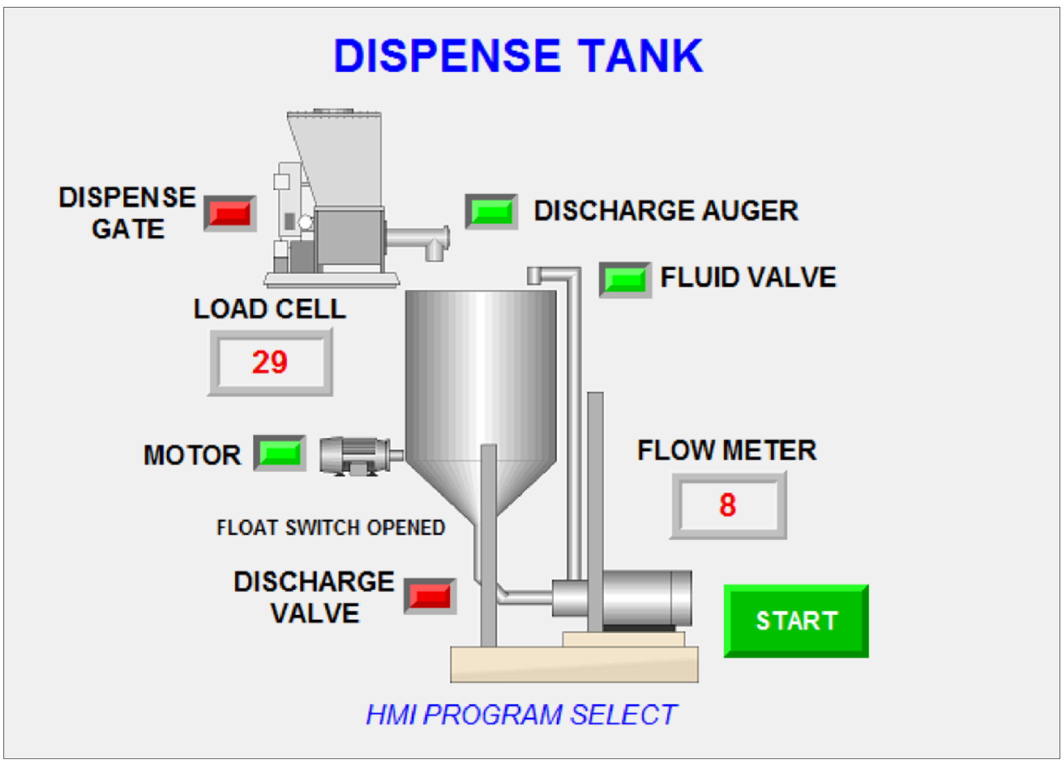
After the timer for the MIXER_MOTOR times out, the MIXER_MOTOR turns OFF and the DISCHARGE_VALUE turn ON.

The LOW LEVEL FLOAT SWITCH will signal the DISCHARGE_VALVE to turn OFF.

NOTES ON RUNNING THE HMI:

Press the START pushbutton.

The process outlined above will be shown on the scaled values and indicators.



REMEMBER TO PRINT TO OneNote FINAL FOLDER

See desktop document <u>Printing Add-On to OneNote</u>

STUDIO 5000 APPLICATION - TORQUE MOTORS

REMEMBER TO PRINT TO OneNote FINAL FOLDER

STUDIO 5000 PROGRAM NAME: TORQUE_FINAL

The logic from the TORQUE lab version is provided.

DO NOT open and use the lab assignment file TORQUE, changes have been made to this file for the final.

SEQUENCE OF OPERATIONS - THE TEXT IN **BOLD** IS THE ADDED FUNCTION.

The operator places a nut onto four bolts and moves the part into a fixture.

The operator will then press and hold the two momentary normally open Start pushbuttons which energize the Down Solenoid. The pushbuttons are pressed for the entire operation. The Down Solenoid is energized and the nut runner is lowered. *Turning off the down solenoid will raise the nut runner.*

When the nut runner travel hits the Down Limit switch, the air motors are energized and will automatically tighten the nuts.

A Torque Sensor is attached to each nut runner.

- **If the torque exceeds 50 foot pounds within two seconds (2) of starting a nut is cross-threaded on the bolt.**
  - **An internal bit CROSS_THRD_x for the nut runner motor that is cross-threaded will be set.**
  - **The internal bit CROSS_THRD_x will not be reset until both START pushbuttons are released.**
  - **An error output, CRS_THD_ERROR, will be turned on.**
  - **When a cross-thread is detected, the nut runner motors and the Down Solenoid will turn off.**

- When the torque reaches 95 to 100 foot pounds, the individual motor will turn off.

- A time out is included, so that after a total of 20 seconds of operations, if torque is not reached by a nut runner motor. The following will occur:
  - The nut runner motors, and the Down Solenoid will turn off.
  - An internal bit for the nut runner motor not at torque will be set.
  - An error output, TO_ERROR, will be turned on.
  - **An Output is turned On to Indicate which bolt has not reached torque**

- When all four motors are de-energized, the Down Solenoid is de-energized, and the nut runner returns to the up position.

The releasing of either START pushbutton will turn off the Down Solenoid and the nut runner motors. The last torque value is retained by the torque sensor.

When both START pushbuttons are released, it will reset the time out error or the cross thread error.

- The torque sensors are reset by the torque sensor transducer.

The upper and lower limits for the torque are defined in tags and are not hard coded constants.

I/O CONFIGURATION:

Slot 2 - STUDIO Emulate 5000 Controller
Slot 3 - 1756-IB16
Slot 4 - 1756-OB16E
Slot 5 - 1756-IF8

NOTE: All Inputs and Outputs, except analog input, are BOOLEAN Data Type shown in a DECIMAL Style and are to be defined as controller tags. The Analog Inputs are REAL Data Type and controller tags. The UPPER_LIMIT and LOWER_LIMIT are DINT Data Type and controller tags. All other tags are to be program tags.

NOTE: For emulation to function properly, Program and Controller Tags MUST HAVE the tagnames listed below.

NOTE: Tags which begin with **em***Tagname* are for the emulation model and are not part of the program.

NOTE: The scaling is performed by the analog input module, the instructions for scaling are not required. Data from the module AIN_1, AIN_2, AIN_3, and AIN_4 can be used directly in the program's logic.

LISTINGS IN **BOLD** ARE THE ADDED CROSS-THREAD TAGS.

INPUTS:

Address	Controller Tag	Description
Local:3:I.Data.10	DN_LS	Down Travel N. O. Limit Switch
Local:3:I.Data.14	START_NO_1	START No. 1 Pushbutton
Local:3:I.Data.15	START_NO_2	START No. 2 Pushbutton

OUTPUTS:

Address	Controller Tag	Description
Local:4:O.Data.0	DN_SOL	Down Solenoid
Local:4:O.Data.1	MOTOR_1	Nut Runner No. 1, Air Motor
Local:4:O.Data.2	MOTOR_2	Nut Runner No. 2, Air Motor
Local:4:O.Data.3	MOTOR_3	Nut Runner No. 3, Air Motor
Local:4:O.Data.4	MOTOR_4	Nut Runner No. 4, Air Motor
Local:4:O.Data.10	TO_ERROR	Time Out Indicator
Local:4:O.Data.11	**CRS_THRD_ERROR**	**Cross Thread Indicator**

ANALOG INPUTS:

Address	Controller Tag	Description
Local:5:I.Ch0Data	AIN_1	Channel 1, Torque Sensor No. 1
Local:5:I.Ch1Data	AIN_2	Channel 2, Torque Sensor No. 2
Local:5:I.Ch2Data	AIN_3	Channel 3, Torque Sensor No. 3
Local:5:I.Ch3Data	AIN_4	Channel 4, Torque Sensor No. 4

INTERNALS:

Controller Tag	Data Type	Description
UPPER_LIMIT	DINT	Upper Torque Setting
LOWER_LIMIT	DINT	Lower Torque Setting
CROSS_THRD_LIMIT	**DINT**	**Cross Thread Torque Setting**

Param/Local Tag	Data Type	Description
CROSS_THRD_1	**Boolean**	**No. 1 Bolt Cross Threaded**
CROSS_THRD_2	**Boolean**	**No. 2 Bolt Cross Threaded**
CROSS_THRD_3	**Boolean**	**No. 3 Bolt Cross Threaded**
CROSS_THRD_4	**Boolean**	**No. 4 Bolt Cross Threaded**
TORQUE_1_NOT_SET	**Boolean**	**No. 1 Bolt Not At Torque**

TORQUE_2_NOT_SET	**Boolean**	**No. 2 Bolt Not At Torque**
TORQUE_3_NOT_SET	**Boolean**	**No. 3 Bolt Not At Torque**
TORQUE_4_NOT_SET	**Boolean**	**No. 4 Bolt Not At Torque**
TORQUE_1_SET	Boolean	No. 1 Bolt Torque Set
TORQUE_2_SET	Boolean	No. 2 Bolt Torque Set
TORQUE_3_SET	Boolean	No. 3 Bolt Torque Set
TORQUE_4_SET	Boolean	No. 4 Bolt Torque Set

TIMERS:

(1) The tags for the status bits DN, TT, and EN are created automatically when timer tag is created.

Param/Local Tag	Data Type	Description
CRS_THRD_DWELL	**Timer**	**Cross Thread Dwell**
CRS_THRD_DWELL.DN (1)	**Boolean**	**Set Cross Thread Error (DN)**
CRS_THRD_DWELL.TT (1)	**Boolean**	**Cross Thread Elapsing (TT) *optional***
CRS_THRD_DWELL.EN (1)	**Boolean**	**Cross Thread Enabled (EN) *optional***
TO_DWELL	Timer	Time Out Dwell
TO_DWELL.DN (1)	Boolean	Set Time Out Error (DN)
TO_DWELL.TT (1)	Boolean	Time Out Elapsing (TT) *optional*
TO_DWELL.EN (1)	Boolean	Time Out Enabled (EN) *optional*

NOTES ON RUNNING THE HMI:

See example HMI screens on next pages.

- The following pattern will be repeated each time the torque motors are reset:

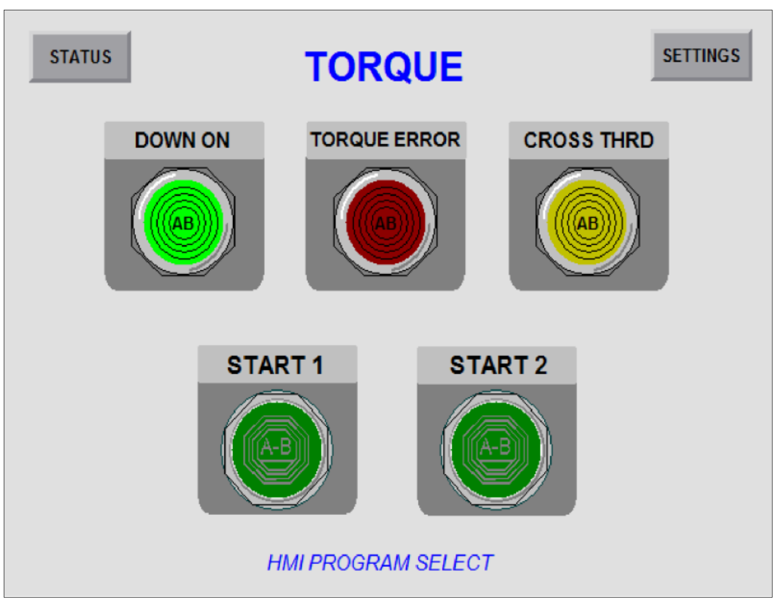
Part 1 - All motors reach torque.
Part 2 - All motors reach torque.
Part 3 - Motor 1 fails to reach torque.
Part 4 - All motors reach torque.
Part 5 - Motor 4 has cross-threaded nut.

- Set the upper and lower torque limits in the Controller Tag Monitor screen to LOWER_LIMIT of 95 and UPPER_LIMIT of 100.
- Place the cursor on the START No. 1 pushbutton and left click.
- Place the cursor on the START No. 2 pushbutton and left click.
- The outputs for the down solenoid should turn on.
- Next the limit switch for the down travel will be set and the nut runner motors should turn on.
- AIN_1, AIN_2, AIN_3, and AIN_4 will be set by the environment to simulate the torque signals values from the individual nut runner motor.
- When all four nut runners have reach torque, or timeout, the down solenoid will turn off.
- **Remember** to release both the START pushbuttons by putting the cursor on the instructions and left clicking for each cycle.

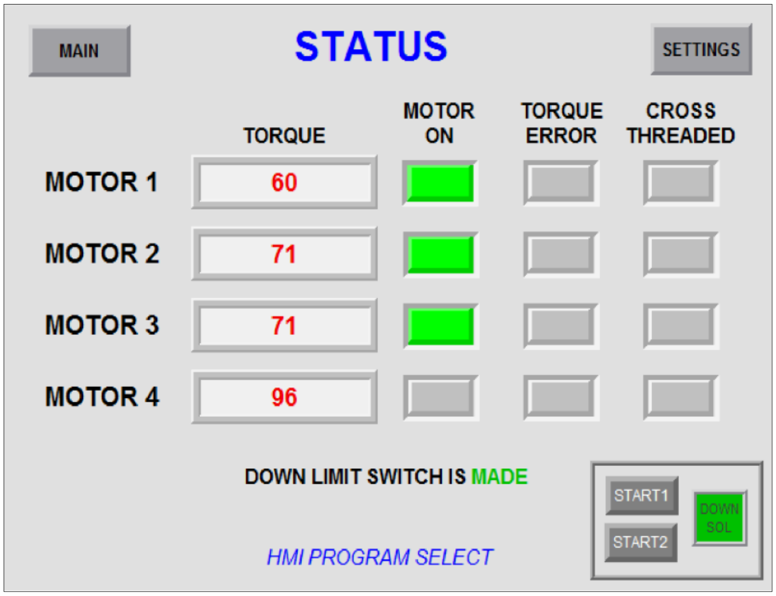
HMI SCREENS

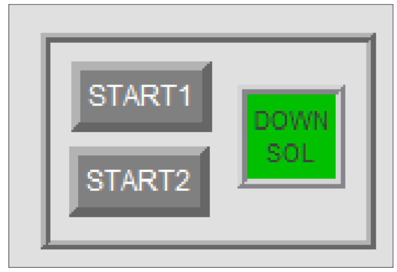
When testing you may use the STATUS screen. *See next page*.

Main Operator Screen



Status Screen - Motors running and Motor 4 at torque

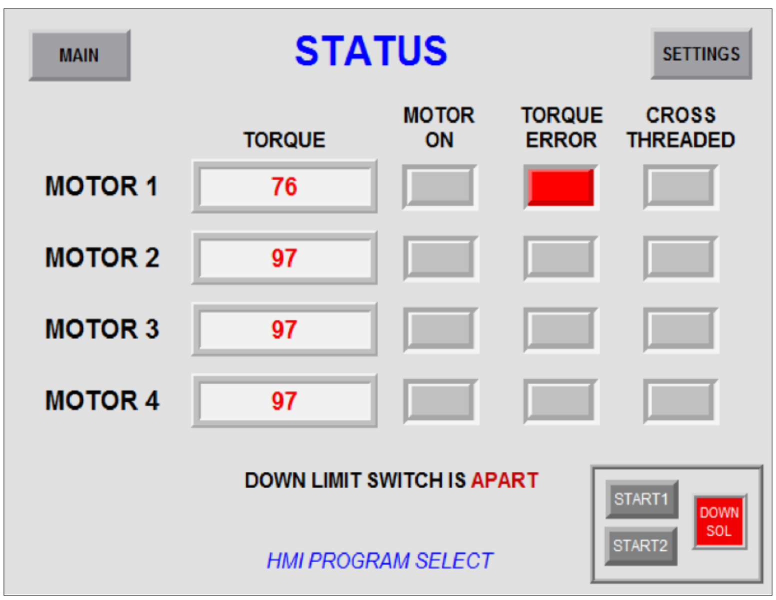




Notice the duplicate START buttons and indicator.

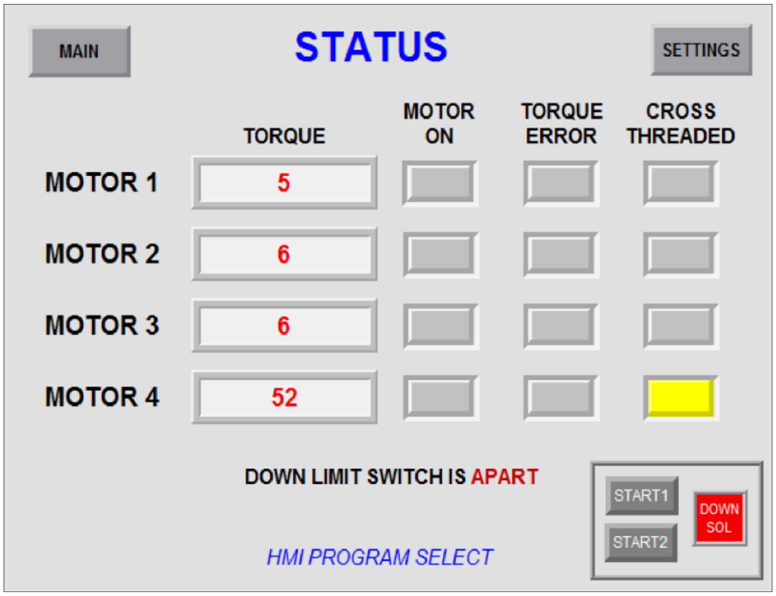
Included for testing purposes. It is *highly advised* you return to the **MAIN** screen on **Torque Error** or **Cross Thread Error** to verify those indicators are set.

Status Screen - Motor 1 Torque Error



(Next Page)

Status Screen - Motor 4 Cross Thread Error



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SIEMENS APPLICATION - PRESS JAM

REMEMBER TO PRINT TO OneNote FINAL FOLDER

SIEMENS PROGRAM NAME: JAM

The rungs outputs are the internal Binary bits (BOOL) used to signal a jam.

JAM NO PART must be set when no part has been detected and the cam-flywheel is back at top-dead-center.

Create the section of control logic that will detect if a part has been ejected from a stamping press. The program monitors a normally open switch that is mounted on a cam which is part of the clutch-brake assembly on the press. When the cam-flywheel returns to top-dead-center and no part was detected, a jam is signaled.

SEQUENCE OF OPERATIONS

The cam-flywheel starts at top-dead-center and the switch, CAM SWITCH, is in the normal position.

1. The AUTO/MAN switch is set to AUTO, voltage applied to AUTO/MAN, and the process will begin. As the cam-flywheel begins to turn, the switch, CAM SWITCH, is actuated. The movement sets the Binary output CAM IN CYCLE. This Binary Output will remain On until CAM CYCLE DONE is set.

2. When the cam-flywheel is back at top-dead-center, CAM CYCLE DONE, and if no part has been detected, PART DETECTED, this is the signal to disengage the flywheel and stop press motion. The Binary output, JAM NO PART, is turned On and remains On. The Binary output, JAM NO PART, is turned off when the AUTO/MAN switch, is turned to the the MAN position, no voltage at the input.

3. While the CAM IN CYLCE is active, after stamping the part the dies separate and ejector pin pushes the stamped part down a discharge chute and past a normally open sensor, PART DETECT, that detects the part. A detected part turns on Binary output, PART DETECTED and remains On until the CAM CYCLE DONE is set.

4. The Binary output CYCLE DONE is set when the cam cycle is complete. The CAM CYCLE DONE and the CAM SWITCH detect the return to top-dead-center.

PLC Tags

INPUTS:

Address	PLC IO Tag	Comments
I0.0	AUTO_MAN	Auto/Man switch, Auto - closed, Man - opened
I0.1	CAM_SWITCH	Normally open cam switch on press fly-wheel
I0.2	PART_DETECT	Part detect ejector sensor for stamping clear press

Create the Global Data Block.

Name of Data Block: Data_block_Press

DATA BLOCK:

Data Block Tag	Data Type	Comments
CAM_IN_CYCLE	BOOL	Press is not at top dead center
JAM_NO_PART	BOOL	No part has been detected by the sensor
PART_DETECTED	BOOL	Part has been detected
CAM_CYCLE_DONE	BOOL	Press is at top dead center cycle is done

OUTPUT ORDER

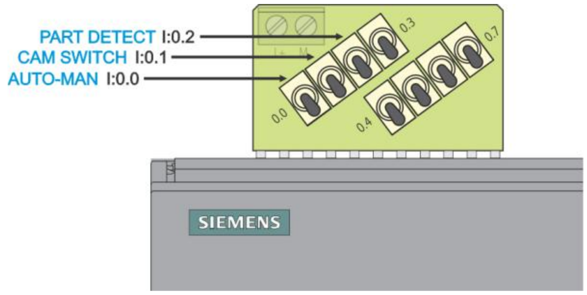
Network 1: “Data_block_Press”.CAM_IN_CYCLE

Network 2: “Data_block_Press”.JAM_NO_PART

Network 3: “Data_block_Press”.PART_DETECTED

Network 4: “Data_block_Press”.CAM_CYCLE_DONE

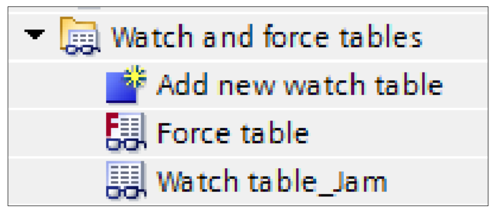
NOTES ON RUNNING:



THERE IS NO HMI

Use the Switches to run the program.

A **Watch Window** has been created for the Boolean Tags in the Data Block.

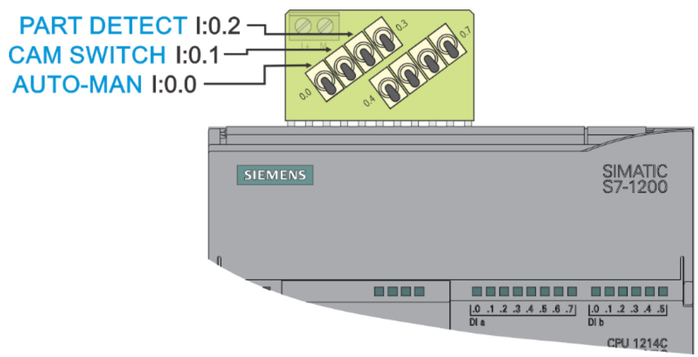


Use the Watch Window's **Float** icon.

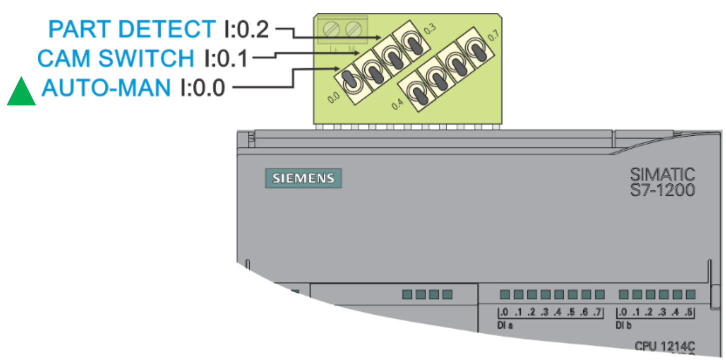
Then open the Main [OB1] and the

<u>INITIAL SETTINGS</u>

All switches DOWN.



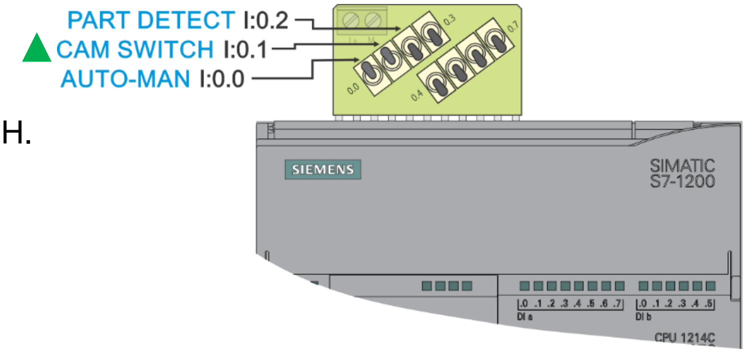
<u>START</u>



The MANUAL/AUTO is set to AUTO to begin the Press, and the cam will be turning.

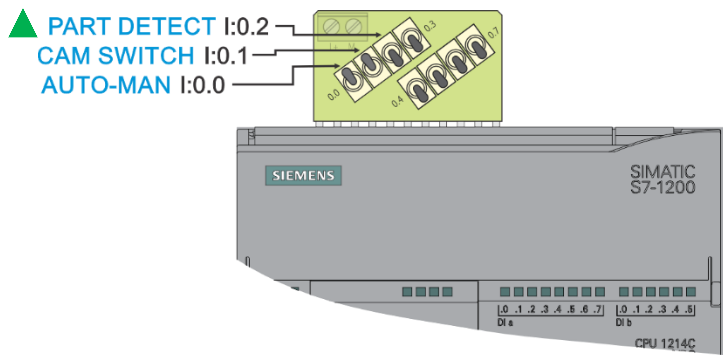
Turn Switch I0.0 ▲ UP.

NO PART JAM SEQUENCE



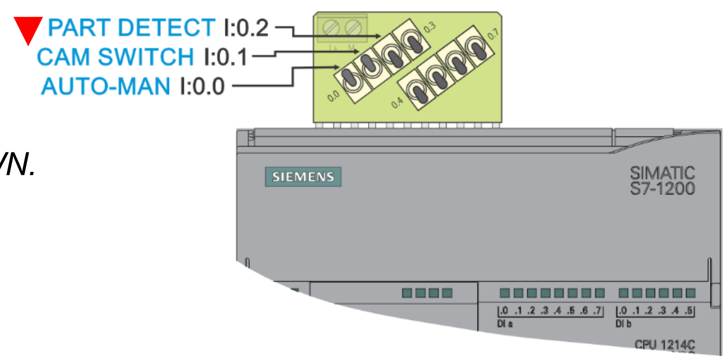
Cam moves from the CAM_SWITCH.

Turn Switch I0.1 ▲ UP.

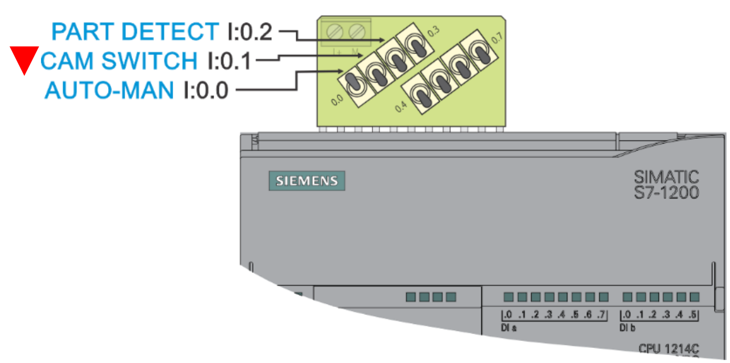


Then, simulate a part moving past the part detect into a hopper.

Turn Switch I0.2 ▲ UP



Then, turn Switch I0.2 ▼ DOWN.



Cam moves into the CAM_SWITCH.

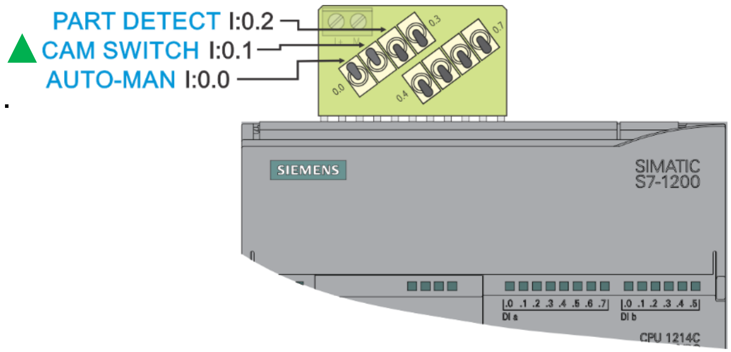
Turn Switch I0.1 ▼ Down.

PART JAM SEQUENCE

Cam moves from the CAM_SWITCH.

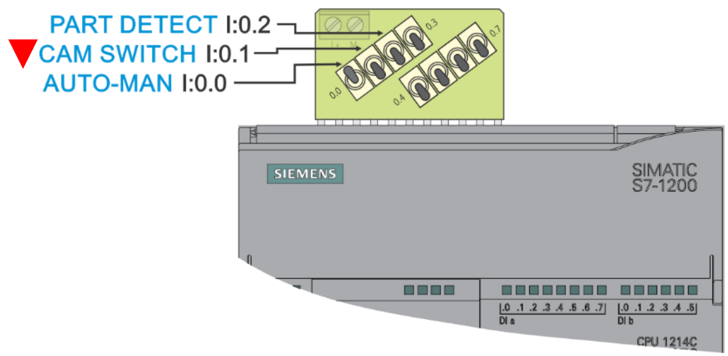
Turn Switch I0.1 ▲ UP.

Press is in cycle.



Cam moves into the CAM_SWITCH without the part being detected.

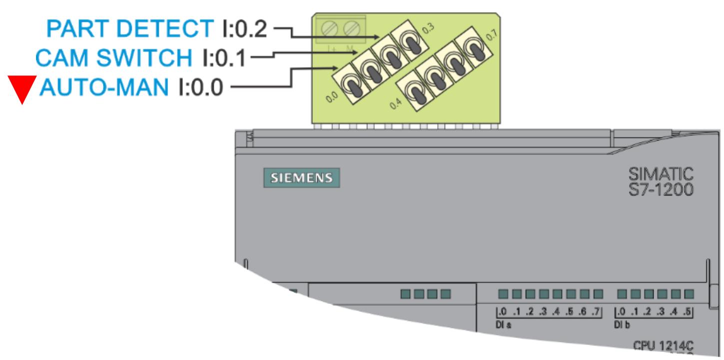
Turn Switch I0.1 ▼ Down.



CLEAR PART JAM SIGNAL

The MANUAL/AUTO is reset to MAN to turn off jam signal.

Turn Switch I0.0 ▼ Down.



REMEMBER TO PRINT TO OneNote FINAL FOLDER

TORQUE HMI

DO NOT PRINT, HMI WILL BE CHECKED ON THE COMPUTER

FactoryTalk View ME PROGRAM NAME: TORQUE FINAL *session*
Open and use the Existing Project. You **DO NOT** need to make a copy or rename.

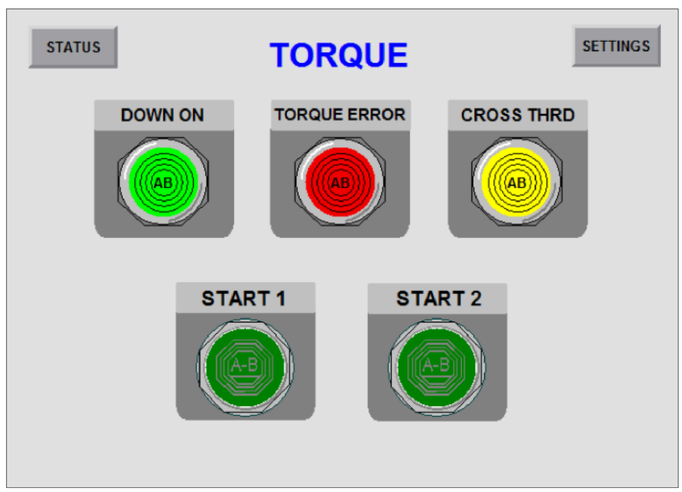
SEE PAGES 24 FOR THE OBJECT EXPLORER AND PAGE 25 FOR THE PROPERTIES PANEL.

IMPORTANT READ

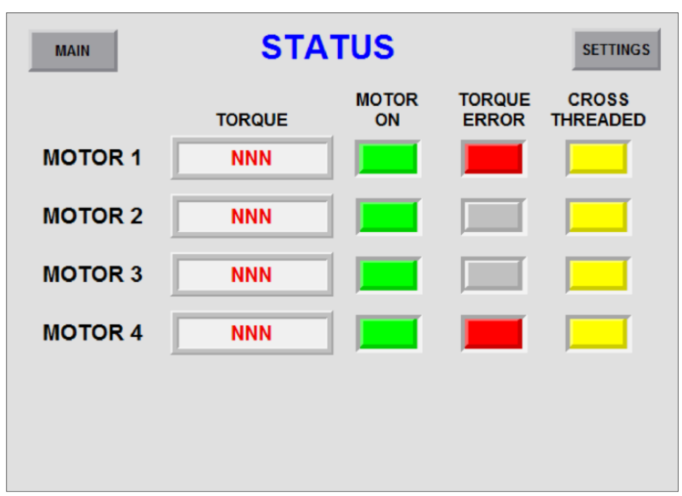
Create a **Communications Shortcut** and assignments for the OFFLINE TAG FILE FOR DEVELOPMENT and DESIGN FOR TESTING. *Use the shortcut name* **FINAL**.

Use your STUDIO 5000 program **TORQUE_FINAL**.

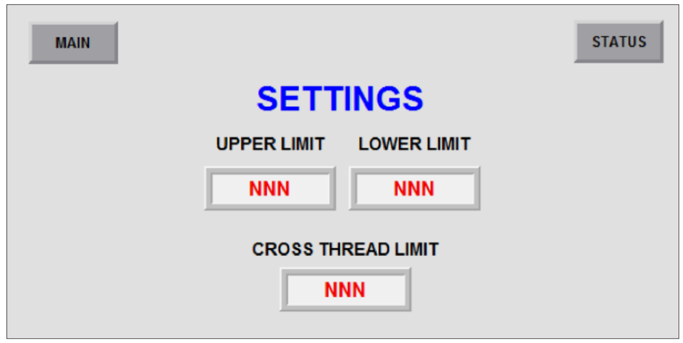
Panel 1 - Main Completed



Panel 2 - Status Completed

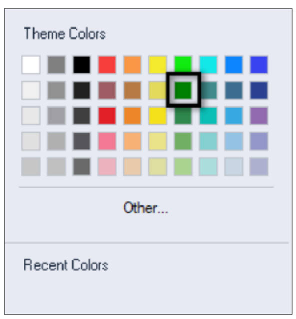


Panel 3 - Settings Completed

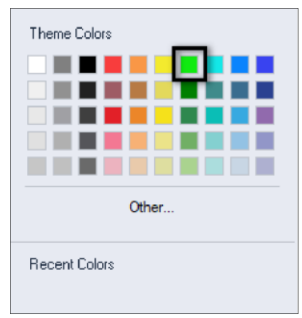


COLORS

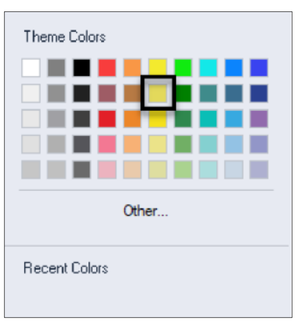
Dark Green



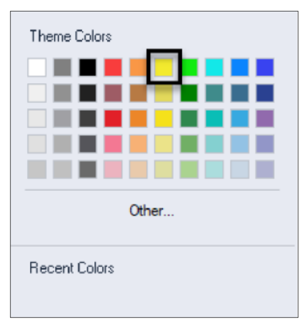
Bright Green



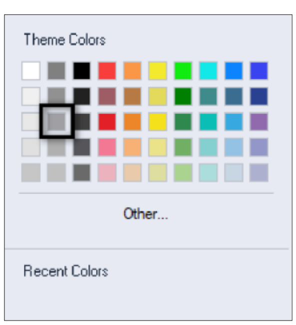
Dark Yellow



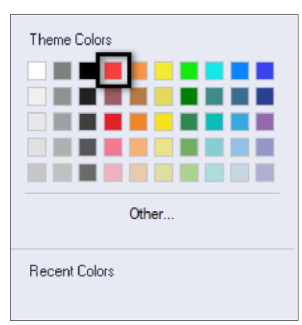
Bright Yellow



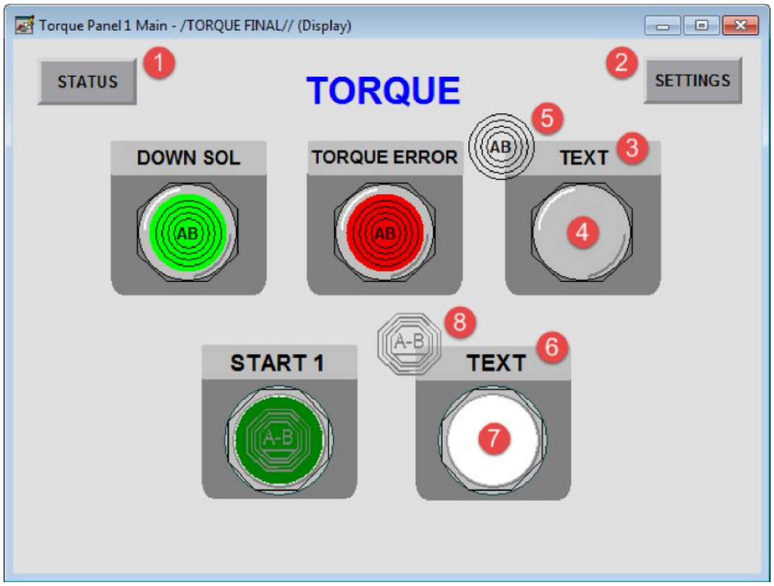
Medium Grey



Bright Red



Panel 1 - Main Modify Requirements



① Goto Screen Button

- Set Screen to Torque Panel 2 - Status

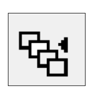
② Goto Screen Button

- Set Screen to Torque Panel 3 - Settings

③ Text

- Change Plate Text CROSS THRD

④ Multistate Indicator Cross Thread

- Create the Multistate Indicator
  Use the Multistate Indicator icon. 

- General
  Border Style: Line
  Back Style: Solid
  Shape: Circle
  Number of States: 2
- Common
  Width: 66
  Height: 66

- States
  - State 0: Dark Yellow
  - State1: Bright Yellow

See page 19 for Color Palettes

- Connections
  - Assign connection to associated CROSS THREAD tag

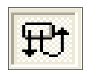
⑤ Set the order for the embossed circle logo to front and drag over the indicator.

⑥ <u>Text</u>

- Change Plate Text START 2

⑦ <u>Maintained Pushbutton Start 2</u>

- <u>Create</u> the Maintained Pushbutton
  Use the Maintained Pushbuttons icon.



- General
  - Border Style: Line
  - Back Style: Solid
  - Shape: Circle
- States
  - State0: Bright Green
  - State1: Dark Green

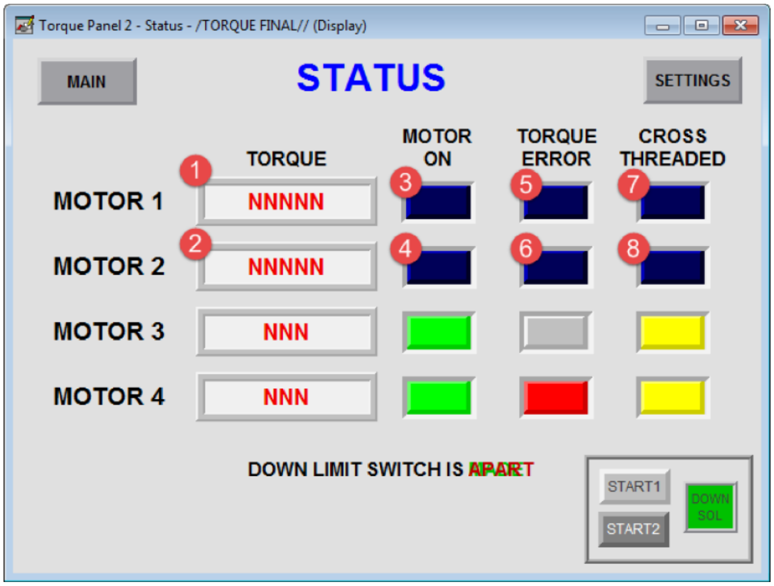
See page 19 for Color Palettes

- Common
  - Width: 74
  - Height: 74
- Connections
  - Assign connection to associated START pushbutton tag.

⑧ Set the order for the embossed octagon logo to front and drag over pushbutton.

(See next page for Panel 2)

Panel 2 - Status Modify Requirements



① Numeric Display

- General: Number of digits to display: 3

- Connections: Assign to associated AIN_x tag.

② Numeric Display

- General: Number of digits to display: 3
- Connections: Assign to associated AIN_x tag.

Note: The Multistate Indicators have a Panel boarder, make sure to select the indicator. It may take two left clicks. The "x' is the corresponding number tag.

③ ④ Multistate Indicators for Motor On

- General: Number of States: 2
- States: State 0 back Color: Meduim Grey
- States: State 1 back Color: Bright Green

See page 19 for Color Palettes

- Assign connection to corresponding MOTOR_x tags.

⑤ ⑥ Multistate Indicators for Torque Error

- General: Number of States: 2
- States: State 0 back Color: Medium Grey

See page 19 for Color Palettes

- States: State 1 back Color: Bright Red

- Connections: Assign to corresponding TORQUE_x_NOT_SET tags.

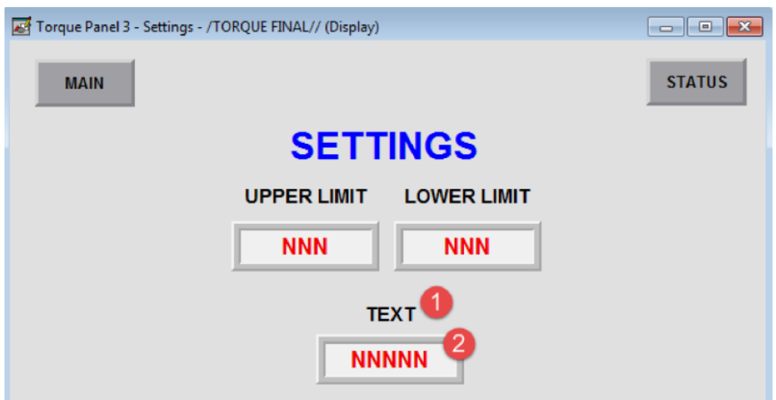
⑦ ⑧ Multistate Indicators for Cross Thread Error

- General: Number of States: 2
- States: State 0 back Color: Medium Grey
- States: State 1 back Color: Bright Yellow

See page 19 for Color Palettes

- Connections: Assign to corresponding CROSS_THRD_x tags.

Panel 3 - Settings Modify Requirements



① Text

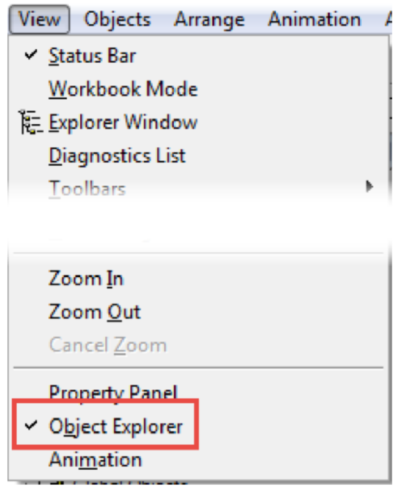
- Change text to CROSS THREAD LIMIT

② Numeric Input for Cross Thread Limit

- Numeric: Minimum value: 0
- Numeric: Maximum value: 150
- Numeric: Number of digits to display: 3

- Connection: Assign to associated CROSS THREAD tag.

OBJECT EXPLORER



Select **View** in the menu and then select **Object Explorer**.

Torque Panel 1 Main

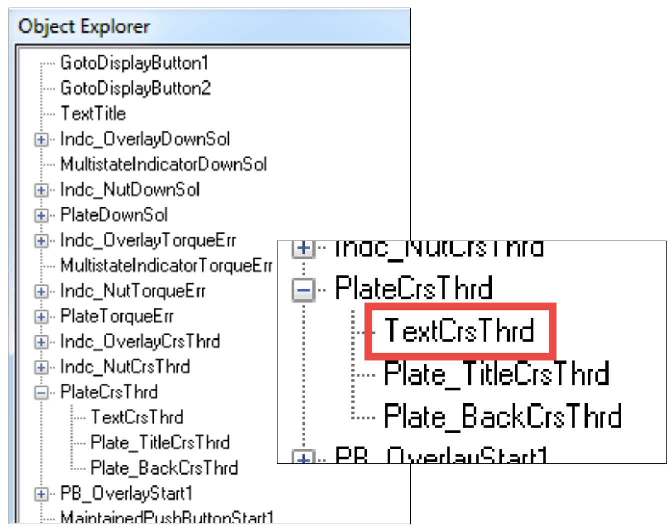


Plate Text

Expand **PlateCrsThrd** and then double left click on **TextCrsThrd** to open the object's Properties dialog.

For Start 2, expand **PlateStart2** and then double left click on **TextStart2** to open the object's Properties dialog.

Torque Panel 2 - Status

For the Multistate Indicators, double left click on the named Indicator to open the object's Properties dialog.

Indicators The "x' is the corresponding number.

Motor ON	**MultistateIndicatorOnMotor*x***
Torque Error	**MultistateIndicatorTorqueErr*x***
Cross Thread	**MultistateIndicatorCrsThrd*x***

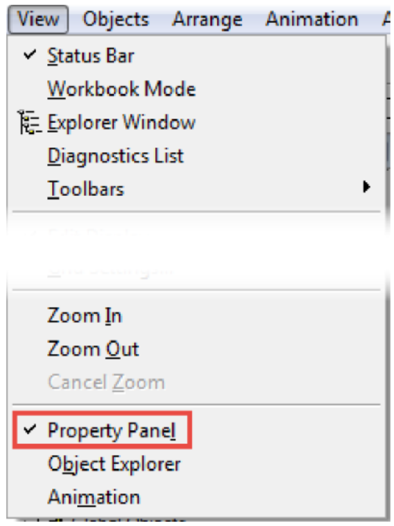
[PROPERTY PANEL]

***NOTE** The Object Explorer and Property Panel can be open at the same time.*

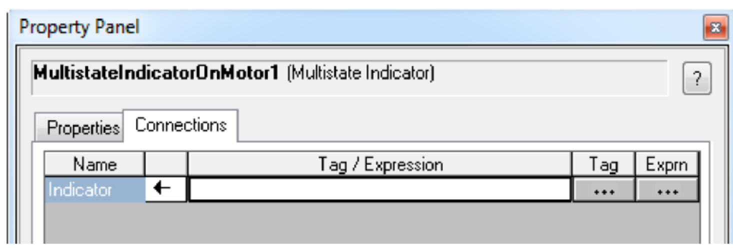
***NOTE** Property Panel remains open when different objects are selected.*

***NOTE** Use Object Explorer to select objects.*

Use the Property Panel to assign or verify tag assignment.



Select **View** in the menu and then select **Property Panel**.



Select the **Connections** tab.

Assign and the tag as normal.